MOBILE RECOMMENDER SYSTEM

KAGGLE DATASET

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01

Main Objective

KAGGLE DATASET

MOBILE DEVICE USAGE AND USER BEHAVIOR DATASET

The dataset facilitates an in-depth analysis of mobile device usage patterns and user behavior. It comprises **700 instances across 11 features.** Ultimately, this data was explicitly compiled to train and evaluate machine learning algorithms. Building upon the creator's original intent, t**his project aims to reframe the mobile device usage data into a recommender system utilizing the following approaches:**

- Content-Based Filtering (utilizing cosine similarity on device feature vectors)
- KNN Collaborative Filtering (leveraging user-user similarity)
- NMF Collaborative Filtering (applying latent factor decomposition)
- Neural Collaborative Filtering (employing an embedding-based MLP)

All models will be compared and evaluated using MAE and RMSE for their rating prediction accuracy, alongside Precision@5 and Recall@5 to assess ranking quality.

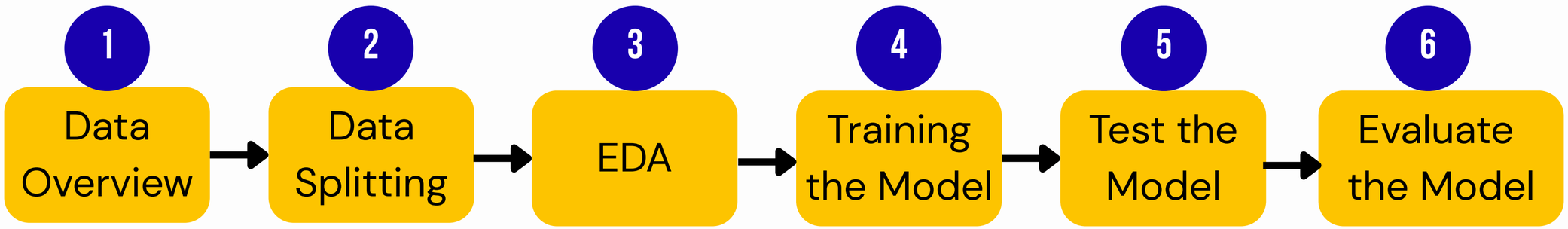
Goals

This study will generate top-N device recommendations per user from each model and determine the optimal algorithm based on the Precision@5 metric.

02

Methodolology and Data Overview

METHODOLOGY



1. Data Overview: Checking the detail and the features of the dataset.
2. Data Splitting: Splitting the data into train and test dataset.
3. EDA: Data cleaning and data visualization of the dataset.
4. Training The Model: Training the 4 models to the training dataset.
5. Test The Model: Test the 4 models to the training dataset.
6. Evaluate The Model: Evaluate all of the models performance.

METHODOLOGY

1 Data Overview

Key Features:

- User ID: Unique identifier for each user.
- Device Model: Model of the user's smartphone.
- Operating System: The OS of the device (iOS or Android).
- App Usage Time: Daily time spent on mobile applications, measured in minutes.
- Screen On Time: Average hours per day the screen is active.
- Battery Drain: Daily battery consumption in mAh.
- Number of Apps Installed: Total apps available on the device.
- Data Usage: Daily mobile data consumption in megabytes.
- Age: Age of the user.
- Gender: Gender of the user (Male or Female).
- User Behavior Class: Classification of user behavior based on usage patterns (1 to 5).

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 700 entries, 0 to 699
Data columns (total 11 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   user_id                     700 non-null    int64
 1   device_model                700 non-null    object
 2   operating_system            700 non-null    object
 3   app_usage_time_min_day_     700 non-null    int64
 4   screen_on_time_hours_day_   700 non-null    float64
 5   battery_drain_mah_day_      700 non-null    int64
 6   number_of_apps_installed    700 non-null    int64
 7   data_usage_mb_day_          700 non-null    int64
 8   age                         700 non-null    int64
 9   gender                      700 non-null    object
 10  user_behavior_class         700 non-null    int64
dtypes: float64(1), int64(7), object(3)
memory usage: 60.3+ KB
```

Datatype of the Features

METHODOLOGY

2 Data Splitting

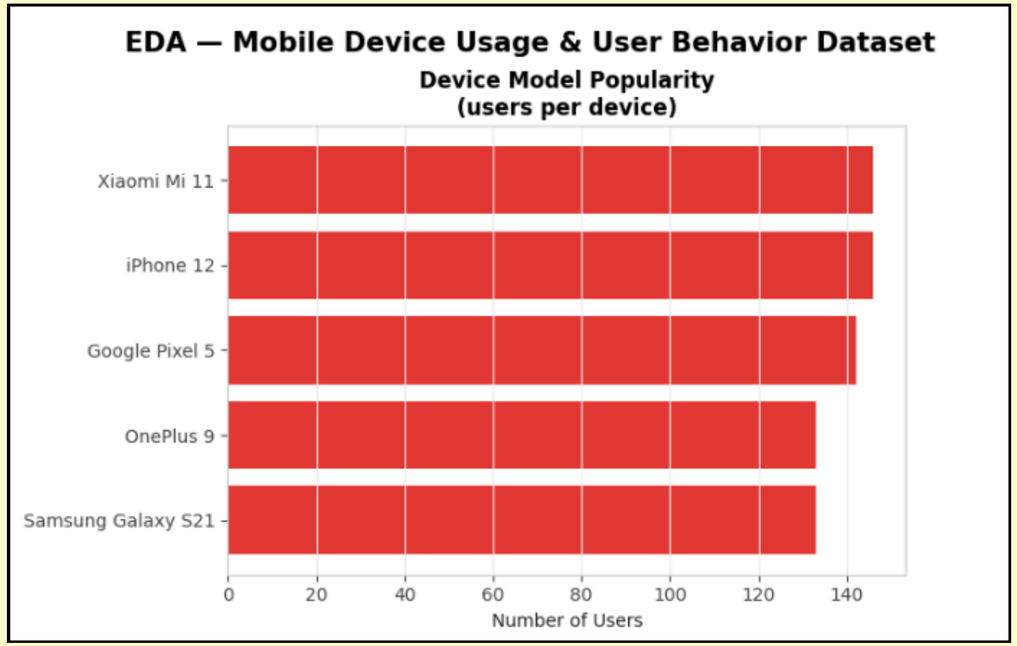
The dataset will be split into 80% for train dataset and 20% for test dataset.

```
Train: 560 ratings | Test: 140 ratings
N_USERS=700 | N_DEVICES=5
```

3 EDA

```
==============================
DATASET OVERVIEW
==============================
  Unique users    : 700
  Unique devices  : 5
  Total ratings   : 700
  Sparsity        : 80.00%
  Avg rating      : 2.99
==============================
```

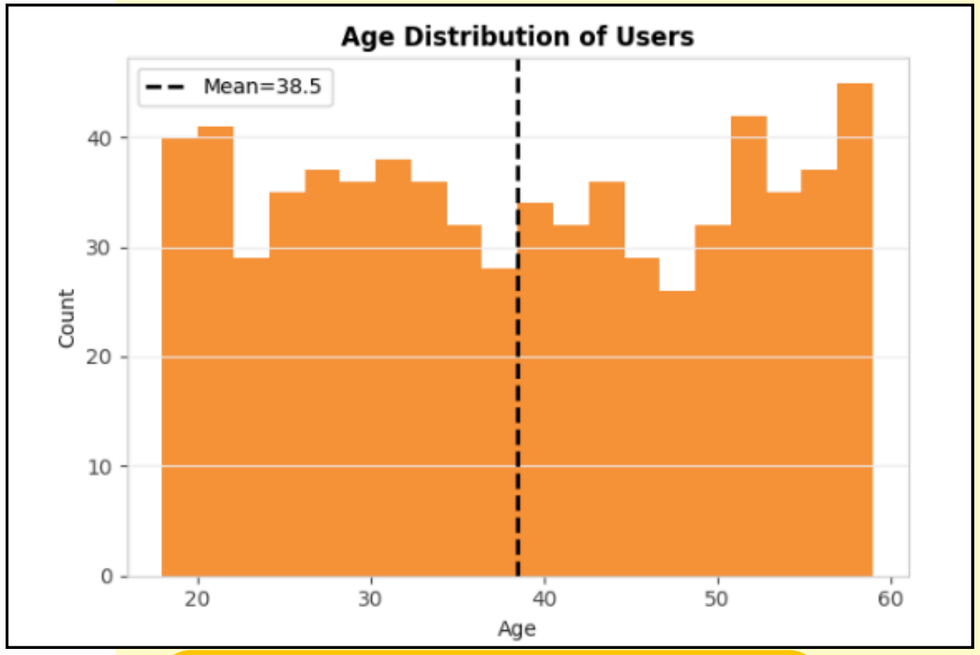
Data Overview



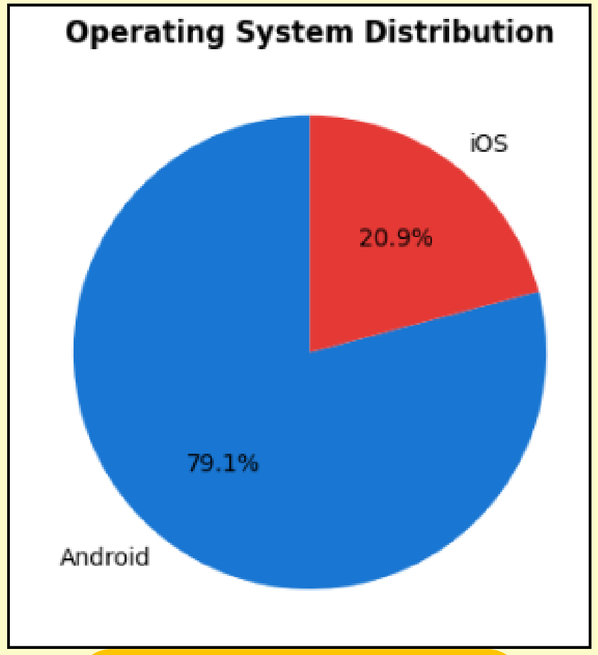
Device Popularity

METHODOLOGY

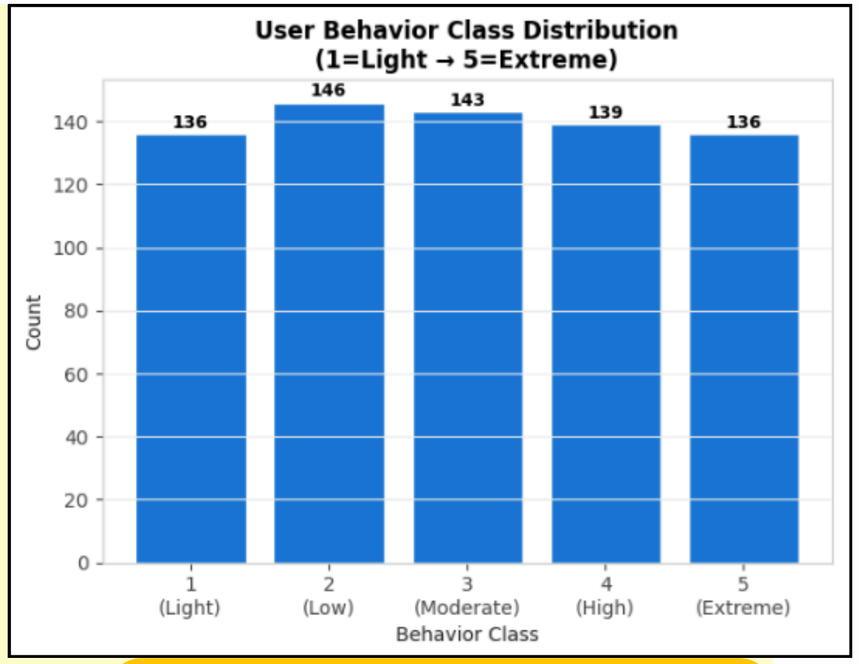
3 EDA



Age Distribution



OS Distribution



User Behavior

- The dataset has an imbalance proportion of OS and Age distributor.
- The popularity for each mobile type and user behavior is almost event.
- A large distribution of age distribution is detected.

03

Model Training, Test, and Evaluation

5 Test and train the Model

CONTENT-BASED FILTERING

Content-Based Filtering works by building a feature vector for each device model based on attributes such as operating system, age group, gender, and behavior bracket.

In this case, **the device feature matrix has a 5 device models are represented across 4 encoded features**.

The model then computes **the cosine similarity between a user's profile and all device vectors to rank recommendations.**

For User 483, the top recommendation is:

- Xiaomi Mi 11 with the highest similarity score of 0.8165
- Samsung Galaxy S21, OnePlus 9, and Google Pixel 5 : all sharing a similarity of 0.6667

```
Device feature matrix shape: (5, 4)
Features: ['os_enc', 'age_group', 'gender_enc', 'behavior_bracket']
                    os_enc  age_group  gender_enc  behavior_bracket
Google Pixel 5         0.0        1.0         1.0               1.0
OnePlus 9              0.0        1.0         1.0               1.0
Samsung Galaxy S21     0.0        1.0         1.0               1.0
Xiaomi Mi 11           0.0        1.0         0.0               1.0
iPhone 12              1.0        1.0         0.0               1.0
```

```
Content-Based Recommendations for user: 483
Device Model                    Similarity
--------------------------------------------
Xiaomi Mi 11                    0.8165
Samsung Galaxy S21              0.6667
OnePlus 9                       0.6667
Google Pixel 5                  0.6667
```

Top-N device model recommendations per user

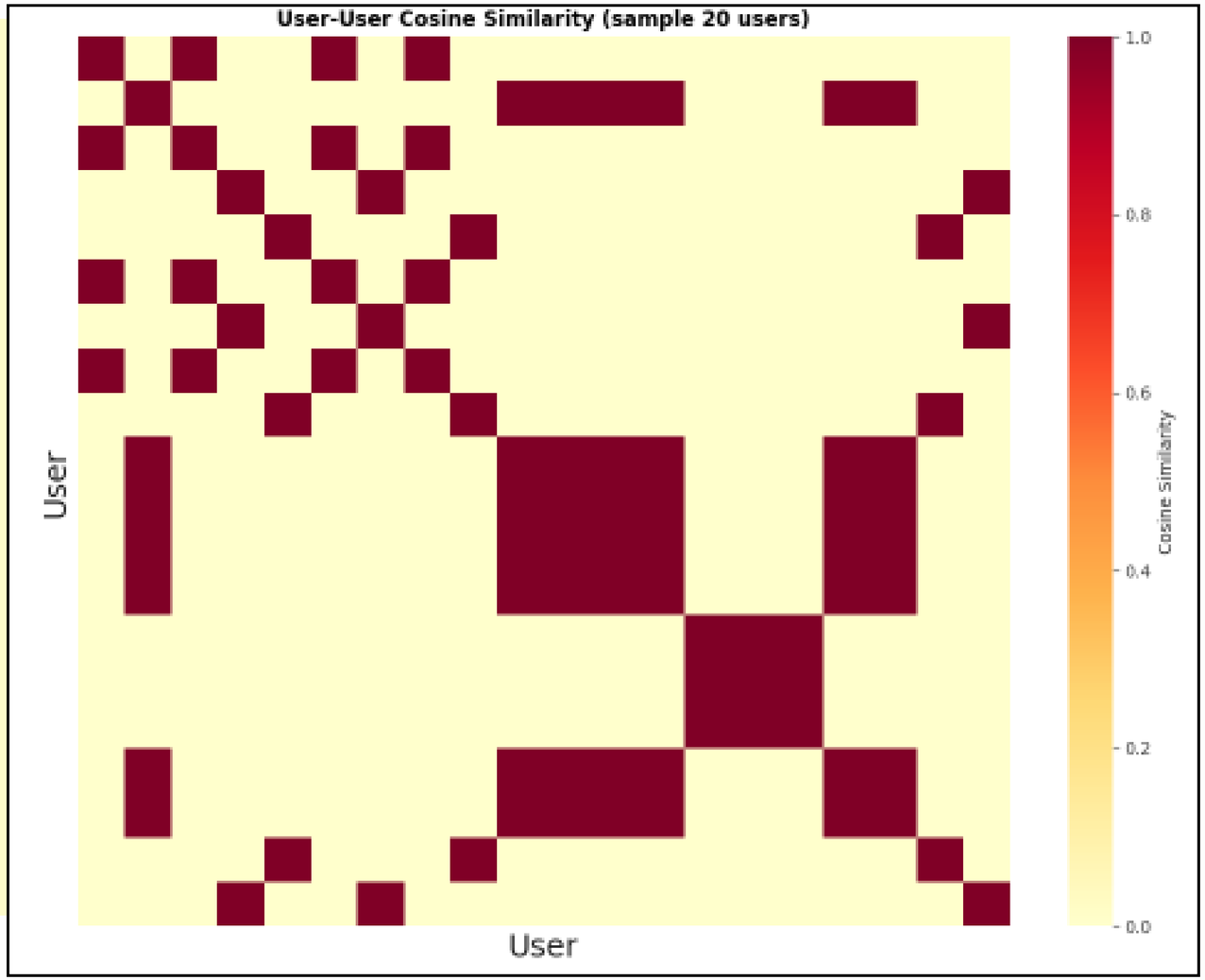
5 Test and train the Model

KNN COLLABORATIVE FILTERING

This heatmap visualizes the **cosine similarity between 20 sampled users** based on their device usage behavior, such as:

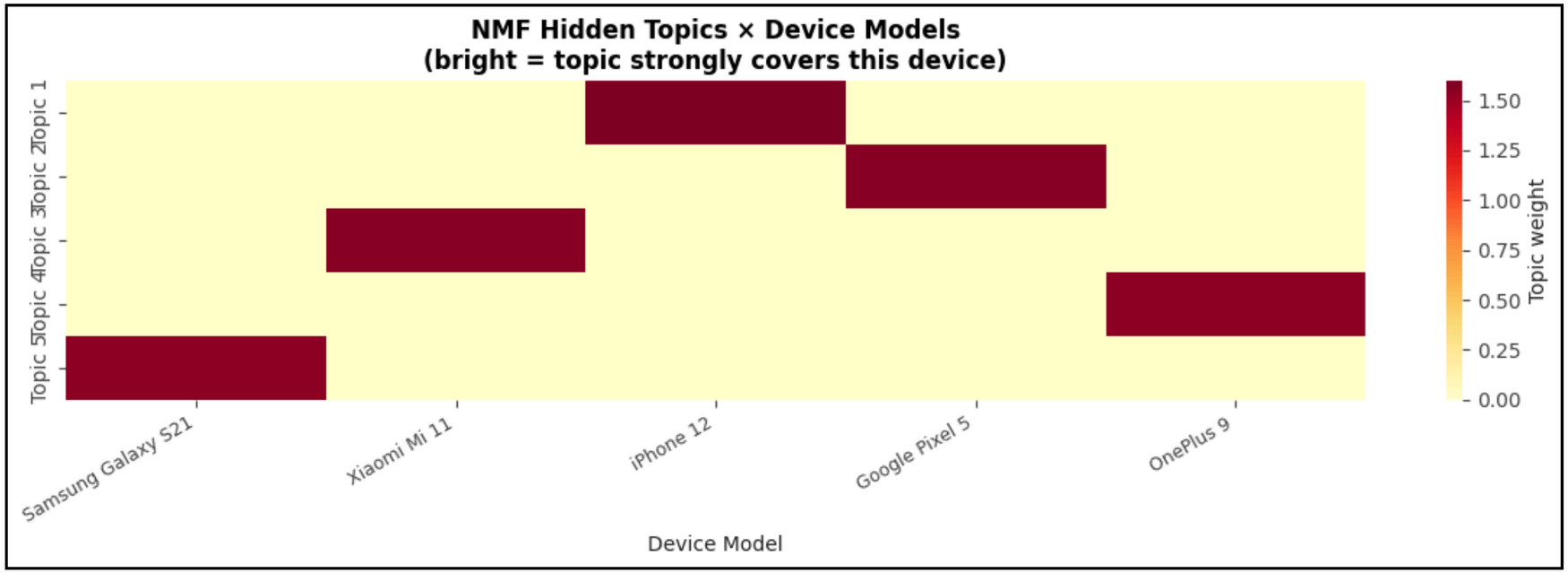
- The diagonal is always 1.0 (a user is perfectly similar to themselves).
- **Dark red blocks** indicate pairs of users with very high similarity meaning they use the same device with the same behavior class.
- P**ale yellow** areas represent near-zero similarity between users who have very different usage profiles.

The highly clustered dark blocks suggest that users naturally group into distinct behavior segments, which is exactly what KNN exploits to find neighbors and generate recommendations.



5 Test and train the Model

NMF COLLABORATIVE FILTERING

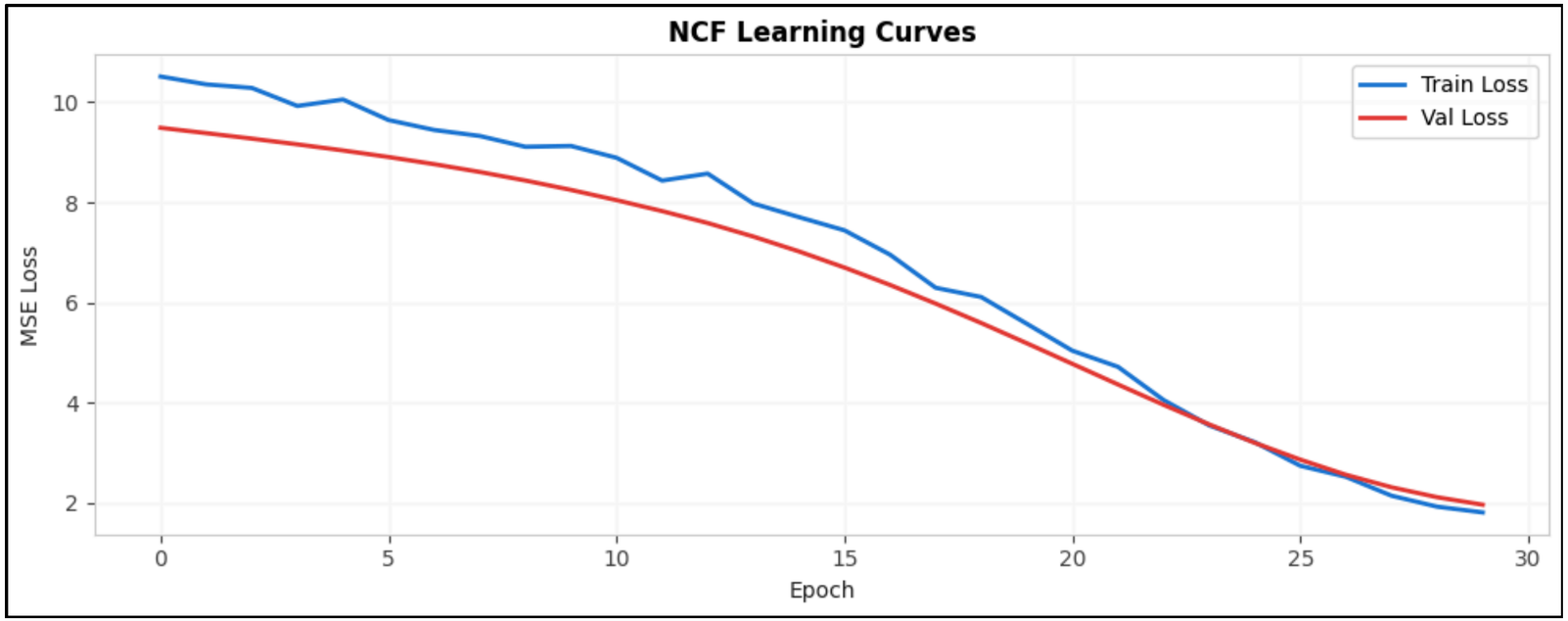


Non-negative Matrix Factorization (NMF) decomposes the user-device matrix into 5 hidden topics, each representing a latent usage archetype (e.g., "heavy data streamers" or "light app users").

The NMF recommendations for User 483 all show a predicted score of 0.0000, which reflects the **challenge of extreme sparsity — with only one interaction per user, NMF struggles to reconstruct meaningful predictions for unseen user-device pairs.**

5 Test and train the Model

NEURAL COLLABORATIVE FILTERING

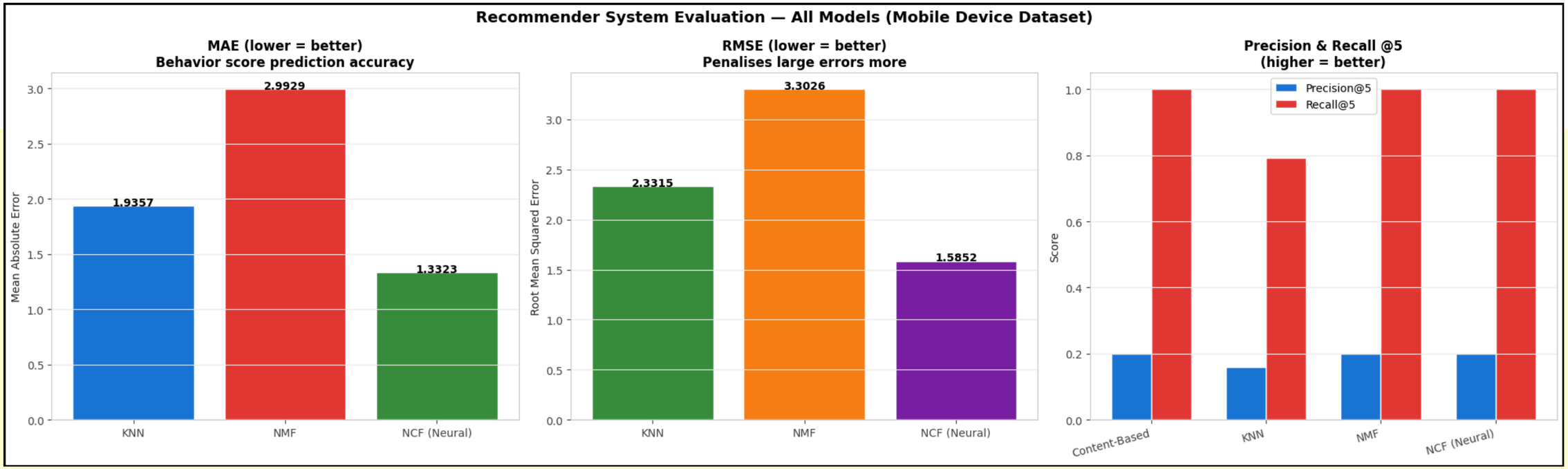


The Neural Collaborative Filtering (NCF) model was trained over 30 epochs with 700 users and 5 devices using an embedding dimension of 16, totaling only 12,369 parameters.

The learning curves show both Train Loss and Validation Loss consistently decreasing from ~10.5 down to ~1.8–1.9, indicating that the model is learning effectively without signs of overfitting — the train and val loss curves remain very close throughout training.

This smooth convergence suggests the NCF model has successfully learned the relationship between user embeddings and device embeddings from the available data.

6 Evaluate the Model

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MODEL EVALUATION RESULTS
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Model	MAE	RMSE	Precision@5	Recall@5
Content-Based	N/A	N/A	0.2	1.0
KNN	1.9357	2.3315	0.1586	0.7929
NMF	2.9929	3.3026	0.2	1.0
NCF (Neural)	1.3323	1.5852	0.2	1.0

- MAE/RMSE : lower is better (behavior score prediction accuracy)
- Precision@5/Recall@5: higher is better (ranking quality)

WINNER (by Precision@5): Content-Based

04 Conclusion and Way Forward

CONCLUSION

We built and compared FOUR recommender system approaches on the Mobile Device Usage dataset (700 users, 5 device models, 700 interactions).

Model	Conclusion
CONTENT-BASED (User Profile + Device Attributes)	Builds a feature vector for each device (OS, dominant age group, gender, behavior bracket) and matches users to similar devices. Works for new users with as little as one interaction. Weakness: limited by how informative the device feature set is.
KNN COLLABORATIVE FILTERING	Finds similar users (by behavior class row) and recommends devices those neighbours used. Simple and interpretable. Struggles on very sparse matrices (1 interaction per user).
NMF COLLABORATIVE FILTERING	Discovers hidden usage archetypes (e.g. 'heavy data streamers', 'light social-media users') and maps both users and devices into these latent topic spaces. Faster at query time than KNN.
NEURAL EMBEDDING (NCF)	Learns user and device fingerprints end-to-end. Most expressive, it can capture non-linear patterns. Best suited a the dataset grows.

WAY FORWARD

CONTENT-BASED

- Add richer device features: screen size, RAM, camera specs
- Weight recent user behaviour more (temporal decay)
- Include App_Usage_Time and Battery_Drain as device profile signals

NMF COLLABORATIVE FILTERING

- Tune n_components (2, 5, 10) using MAE on validation set
- Try SVD or ALS which handle sparse implicit feedback better
- Incorporate side features (age, gender) as auxiliary matrices

KNN COLLABORATIVE FILTERING

- Collect more interactions (e.g. app category ratings per device)
- Try item-based KNN: find similar devices, not similar users
- Use Pearson correlation instead of cosine for bias correction

NEURAL EMBEDDING (NCF)

- Scale embed_dim (16→32→64) as dataset grows
- Add user side features (age, gender, OS) as additional inputs
- Try two-tower architecture for scalable retrieval
- Use BPR loss (Bayesian Personalised Ranking) for implicit data

THANK YOU